Transfer Learning Mini-Lab

CPECOG2 · Deep Neural Networks

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Setup

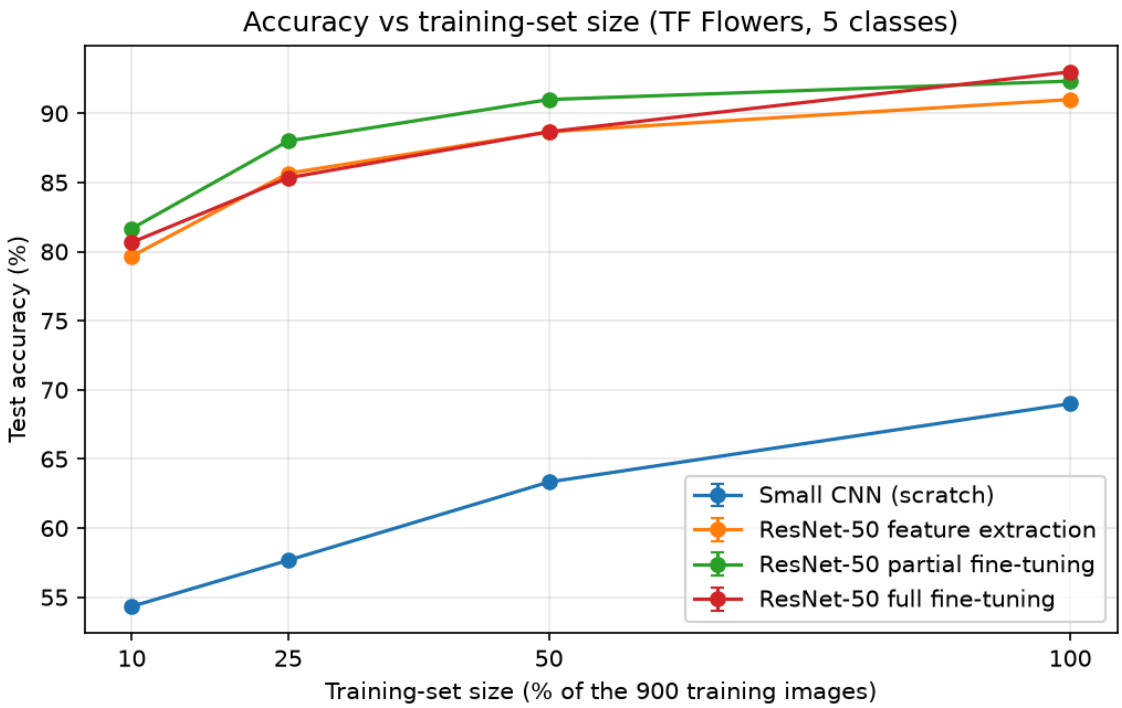
- **Dataset:** TF Flowers, 5 classes (daisy, dandelion, roses, sunflowers, tulips), 300 images per class.
- **Split:** 900 train, 300 validation, 300 test. The same split is used for every run.
- **Models:** a small CNN from scratch, and ResNet-50 (ImageNet weights) with feature extraction, partial fine-tuning, and full fine-tuning.
- **Seed:** one (0). One test image is 0.33 points, so we count gaps of 2 points or less as ties.

Results (100% of training data)

Model	Accuracy	Macro-F1	Trainable params	Time (s)
Small CNN (scratch)	0.690	0.689	94,341	106
Feature extraction	0.910	0.910	10,245	503
Partial fine-tuning	0.923	0.924	14,974,981	112
Full fine-tuning	0.930	0.930	23,518,277	270

Time is the total training time with early stopping. Feature extraction trains the fewest weights but took the longest, because its small head needed 35 epochs and each epoch still runs the full ResNet-50.

Accuracy vs Training-Set Size



Train images	Scratch	Feature extraction	Partial FT	Full FT
90 (10%)	0.543	0.797*	**0.817**	0.807
225 (25%)	0.577	0.857*	**0.880**	0.853
450 (50%)	0.633	0.887	**0.910**	0.887
900 (100%)	0.690	0.910	0.923	**0.930**

* Still improving slowly when it hit the 40-epoch limit, so slightly underestimated.

Winning Strategy

Partial fine-tuning won at 25% and 50%. At 10% and 100%, the three ResNet-50 models were within 2 points of each other, so those sizes are ties. Scratch was last at every size, 24 to 30 points behind. ResNet-50 already learned edges, textures, and shapes from ImageNet, and flowers look like ImageNet photos, so those features carry over. The small CNN has to learn them from a few hundred images. Feature extraction can’t adapt its frozen features, so it trails partial fine-tuning. Partial fine-tuning lets the top layers adapt to flowers and keeps the rest fixed. Full fine-tuning has 23.5 M weights to fit, which is too many for small data, and it caught up only at 100%. This matches the lecture: with small, similar data, fine-tune only the top layers. With more data, unfreezing more layers starts to pay off.